

parameters, like range of the target. By displaying a ready symbology on the sight, it also indicates to the gunner that he can shoot.

Autoloader and shell loading. The firing solution has been derived, the gun is laid, what about the ammunition? Generally, it will be loaded even before deriving the firing solution. Who does it? Western tanks require a soldier known as loader who, under the command of the gunner, picks up the appropriate shell from ammunition rack and loads it in the gun. A slight mistiming can result in amputation for him.

The Soviet Union/Russian-origin tanks use an important system called autoloader. The autoloader is a separate system but loosely interacts with the FCS. The autoloader selects the shell according to the gunner's command and loads it in the gun. During combat, different types of shells are required to tackle various situations. For this, the gunner inputs the re-

quired shell type in his console and the autoloader loads accordingly. The FCS checks whether the autoloader has loaded the shell into the gun or not.

Engagement sequence

In a typical combat, the tank commander (with his panoramic sight) searches for a suitable target, detects it and presses a switch called align switch. The FCS slews the turret to face the target. The gunner then takes over the engagement of the target. The gunner aligns the reticle on the target and then the FCS takes over the engagement from him. The FCS starts performing the sequence of events from ranging and tachometric process (detecting the speed of the target), lays the gun and when all are set, the gunner can fire the gun.

The commander, after having assigned a target to the gunner, uses his panoramic sight to locate the next target. If he has located a target

even before the current firing, it is stored in the memory of the FCS. After firing and damage assessment of the target is over, the gunner can take up this new target from the FCS memory. The commander can keep on looking for new targets through his sight and, if required, he can take over the engagement from the gunner and himself shoot on sudden threats. In the past such an engagement sequence would have been a dream! The modern tanks are said to be capable of typically engaging around six different targets within 35 seconds while on the move.

Apart from these commander's and gunner's sights, the tank driver also has a day/night-capable sight. With that he can drive the tank as efficiently during nights as he does during day; the driver needs not to even switch on the headlights to drive. In some very latest tanks, rear facing cameras feed imagery to the driver who can easily reverse the tank, or manoeuvre into the




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